DEFENCE JOURNEY

# NDA Mock Test 1

Practice test in the NDA pattern with answer key and explanations

**NDA | MOCK 1**

**TIP:** Time: 40 minutes. Questions: 30 (Maths 12, English 8, General Knowledge 10). Marking: +1 for a correct answer, −1/3 for a wrong answer, 0 if left blank. This is a short practice test in the NDA pattern, not the full-length paper. Attempt it without looking at the notes, then check the answer key on the last page.

## Part A: Mathematics

**Q1.** If A = {1, 2, 3, 4} and B = {3, 4, 5, 6}, then n(A ∪ B) is

(A) 6
(B) 8
(C) 5
(D) 4

**Q2.** The value of $i^{25}$ is

(A) 1
(B) −1
(C) i
(D) −i

**Q3.** If α and β are the roots of $x^2 - 5x + 6 = 0$, then $\alpha^2 + \beta^2$ equals

(A) 13
(B) 17
(C) 25
(D) 11

**Q4.** The 10th term of the AP 3, 7, 11, 15, ... is

(A) 47
(B) 35
(C) 43
(D) 39

**Q5.** If ${}^{n}C_{2} = 45$, then n is

(A) 9
(B) 12
(C) 10
(D) 8

**Q6.** The value of the determinant |2 3; 4 5| is

(A) 22
(B) 2
(C) −22
(D) −2

**Q7.** The value of sin 30° + cos 60° is

(A) 1/2
(B) √3/2
(C) 1
(D) √3

**Q8.** If $\tan\theta = 3/4$ and $\theta$ is acute, then $\sin\theta$ is

(A) 3/5
(B) 5/3
(C) 4/5
(D) 3/4

**Q9.** The distance between the points (1, 2) and (4, 6) is

(A) 4
(B) 7
(C) 5
(D) 3

**Q10.** The derivative of $x^3 + 2x$ at $x = 1$ is

(A) 5
(B) 6
(C) 3
(D) 4

**Q11.** The value of $\int_0^1 (2x + 1)\,dx$ is

(A) 3
(B) 2
(C) 4
(D) 1

**Q12.** If $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, then $\mathbf{a} \cdot \mathbf{b}$ is

(A) 3
(B) 1
(C) 2
(D) 4

## Part B: English

**Q13.** Choose the word closest in meaning to ABUNDANT.

(A) Tiny
(B) Costly
(C) Scarce
(D) Plentiful

**Q14.** Choose the word opposite in meaning to BENEVOLENT.

(A) Generous
(B) Friendly
(C) Cruel
(D) Kind

**Q15.** Fill in the blank: He is senior ___ me.

(A) from
(B) with
(C) than
(D) to

**Q16.** Fill in the blank: Neither of the boys ___ present.

(A) have been
(B) was
(C) are
(D) were

**Q17.** What does the idiom 'once in a blue moon' mean?

(A) Very rarely
(B) Every month
(C) At night
(D) Quite suddenly

**Q18.** One word for 'one who loves mankind':

(A) Optimist
(B) Misanthrope
(C) Philanthropist
(D) Patriot

**Q19.** Fill in the blank: She has been living here ___ 2015.

(A) by
(B) since
(C) from
(D) for

**Q20.** Choose the correct passive form of: 'She wrote a letter.'

(A) A letter was written by her.
(B) A letter is written by her.
(C) A letter was wrote by her.
(D) A letter has been written by her.

## Part C: General Knowledge

**Q21.** Who discovered the electron?

(A) Niels Bohr
(B) Rutherford
(C) James Chadwick
(D) J. J. Thomson

**Q22.** The SI unit of power is

(A) Pascal
(B) Watt
(C) Newton
(D) Joule

**Q23.** The First Battle of Panipat (1526) was fought between Babur and

(A) Ibrahim Lodi
(B) Rana Sanga
(C) Sher Shah
(D) Akbar

**Q24.** Which gas is the most abundant in the Earth's atmosphere?

(A) Argon
(B) Carbon dioxide
(C) Nitrogen
(D) Oxygen

**Q25.** The longest river in India is

(A) Ganga
(B) Brahmaputra
(C) Godavari
(D) Yamuna

**Q26.** Who was the first President of India?

(A) Dr. Zakir Husain
(B) Dr. S. Radhakrishnan
(C) Jawaharlal Nehru
(D) Dr. Rajendra Prasad

**Q27.** Deficiency of which vitamin causes scurvy?

(A) Vitamin A
(B) Vitamin D
(C) Vitamin C
(D) Vitamin K

**Q28.** The Tropic of Cancer passes through which of these states?

(A) Rajasthan
(B) Punjab
(C) Kerala
(D) Bihar

**Q29.** The Quit India Movement was launched in the year

(A) 1935
(B) 1946
(C) 1930
(D) 1942

**Q30.** The chemical formula of baking soda is

(A) $CaCO_3$
(B) $NaHCO_3$
(C) $Na_2CO_3$
(D) NaCl

## Answer Key

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| 1 | A | 2 | C | 3 | A | 4 | D | 5 | C |
| 6 | D | 7 | C | 8 | A | 9 | C | 10 | A |
| 11 | B | 12 | A | 13 | D | 14 | C | 15 | D |
| 16 | B | 17 | A | 18 | C | 19 | B | 20 | A |
| 21 | D | 22 | B | 23 | A | 24 | C | 25 | A |
| 26 | D | 27 | C | 28 | A | 29 | D | 30 | B |

## Explanations

**Q1. (A)** $A \cup B = \{1, 2, 3, 4, 5, 6\}$, which has 6 elements.

**Q2. (C)** $25 = 4 \times 6 + 1$, so $i^{25} = (i^4)^6 \cdot i = i$.

**Q3. (A)** $\alpha + \beta = 5$ and $\alpha\beta = 6$. $\alpha^2 + \beta^2 = 25 - 12 = 13$.

**Q4. (D)** $a = 3$, $d = 4$. $T_{10} = 3 + 9 \times 4 = 39$.

**Q5. (C)** $n(n - 1)/2 = 45$, so $n(n - 1) = 90$, giving $n = 10$.

**Q6. (D)** $2 \times 5 - 3 \times 4 = 10 - 12 = -2$.

**Q7. (C)** $1/2 + 1/2 = 1$.

**Q8. (A)** The triangle has sides 3, 4 and hypotenuse 5, so $\sin\theta = 3/5$.

**Q9. (C)** $\sqrt{[(4 - 1)^2 + (6 - 2)^2]} = \sqrt{(9 + 16)} = 5$.

**Q10. (A)** $f'(x) = 3x^2 + 2$, and $f'(1) = 5$.

**Q11. (B)** $[x^2 + x]$ from 0 to 1 $= 1 + 1 = 2$.

**Q12. (A)** $1 \times 2 + 2 \times (-1) + 3 \times 1 = 2 - 2 + 3 = 3$.

**Q13. (D)** Abundant means existing in large quantities, i.e. plentiful.

**Q14. (C)** Benevolent means kind and well-meaning. The opposite is malevolent or cruel.

**Q15. (D)** Senior, junior, superior and inferior are followed by *to*.

**Q16. (B)** *Neither* takes a singular verb. 'Was' is the singular past form.

**Q17. (A)** The idiom means something that happens very seldom.

**Q18. (C)** A philanthropist loves and helps humanity. A misanthrope hates mankind.

**Q19. (B)** *Since* is used with a point in time (2015). *For* is used with a period.

**Q20. (A)** Simple past active becomes was/were + past participle (written).

**Q21. (D)** J. J. Thomson discovered the electron in 1897.

**Q22. (B)** Power = work/time. Its SI unit is the watt (1 joule per second).

**Q23. (A)** Babur defeated Ibrahim Lodi at Panipat in 1526 and founded the Mughal Empire.

**Q24. (C)** Nitrogen makes up about 78% of the air.

**Q25. (A)** The Ganga (about 2,525 km) is the longest river flowing in India.

**Q26. (D)** Dr. Rajendra Prasad served from 1950 to 1962.

**Q27. (C)** Lack of vitamin C (ascorbic acid) causes scurvy.

**Q28. (A)** It passes through Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.

**Q29. (D)** Gandhiji gave the call 'Do or Die' on 8 August 1942 at Bombay.

**Q30. (B)** Baking soda is sodium hydrogen carbonate. $Na_2CO_3$ is washing soda.